PROPOSALS

Mokila Kshitij Reddy
(24IM10045)

General Secretary Students' Welfare

Technology Students' Gymkhana

Indian Institute of Technology Kharagpur

Academic Session : 2026-2027

Proposal 1

Changes in the Life Skill(LS10001) Course on Mental Well-being, Awareness, and Psychometric Screening

1.1 Aim

To transition the institute's mental health ecosystem from a reactive, demand-driven model to a preventive, structured, and awareness-oriented framework by making some changes in the present existing Life Skills course. This initiative aims to ensure early identification of psychological concerns, normalize engagement with counselling services, and build a campus culture that actively supports mental well-being.

1.2 Analysis of Current Scenario

Delayed Identification of Psychological Distress

A significant proportion of students at IIT Kharagpur experience stress, anxiety, and adjustment challenges during their transition. These issues often go undetected, as students tend to seek help only at advanced stages.

Predominantly Reactive Support System

The current counselling framework at IIT Kharagpur relies heavily on self-referral, limiting early engagement. As a result, the system functions more as a crisis-response mechanism than a preventive support structure.

Persistent Stigma and Psychological Barriers

Stigma remains a significant barrier at IIT Kharagpur, with counselling often perceived as a last resort or a sign of weakness by the students. Fear of judgment and lack of familiarity discourage students from seeking help early.

Absence of Structured Screening Mechanism

There is currently no institution-wide baseline assessment of student mental well-being or system to identify at-risk individuals early. This limits the ability to categorize needs and provide targeted interventions.

1.3 Proposed Solution

Major changes in the existing Life Skill course like:

- Digital psychometric screening
- Guided exposure to SETU

1.4 Implementation Plan

Initial Psychometric Screening

Conducted online in CIC, assessing stress and anxiety levels, emotional well-being, coping patterns, and academic and social adjustment

Data-Driven Student Segmentation

Students will be grouped based on their screening responses to ensure meaningful and effective engagement during sessions. This approach enables the formation of groups with similar perspectives and needs, fostering more relevant discussions and participation.

Guided Counselling Center Exposure

Students will participate in SETU sessions within their groups to enable focused discussions, improve relatability, and enhance engagement.

Objectives:

- Familiarize students with:
    - SETU
    - Counsellors
    - Process of seeking help
- Reduce hesitation and fear

- Build early trust and approachability

Stigma Reduction Strategy

Achieved through universal participation, direct exposure to SETU, normalization of mental health discussions, and peer-level awareness.

Data Privacy and Ethical Safeguards

All data will remain strictly confidential, with no misuse and only consent-based sharing.

Monitoring and Evaluation

Annual review based on participation, student feedback, counsellor observations, and help-seeking trends.

1.5 Expected Outcomes

Short-Term

Universal mental health awareness, increased familiarity with SETU, and reduced stigma.

Medium-Term

Improved help-seeking behavior, early identification of students needing support, and a stronger peer-support ecosystem.

Long-Term

Shift toward preventive mental health care, leading to a healthier, more resilient student community with improved academic engagement and overall well-being.

1.6 Conclusion

This proposal creates a scalable framework integrating awareness, assessment, and engagement. It embeds mental health into the academic system, enabling proactive and inclusive support aligned with global best practices.

Proposal 2

Upgrading SETU & ApnaInsti — A Feature Proposal for KGP Students

2.1 Aim

To strengthen SETU and ApnaInsti by incorporating three features that simplify counselling access, support course-based academic interaction, and improve awareness of student opportunities through a centralized calendar, using existing infrastructure and minimal additional resources.

2.2 Analysis of Current Scenario

There are three main issues in the current digital system at IIT Kharagpur that make things harder for students. Booking a counselling session is slow because it requires sending an email and waiting for a reply, which can take time when support is needed quickly. Also, there is no proper platform where students can ask and resolve academic doubts, so they depend on WhatsApp groups or personal messages, which are not always reliable.

In addition, there is no single place where students can find all important opportunities like internships, competitions, or research programs. Information is spread across multiple WhatsApp groups, emails, and individual sources, making it difficult to track consistently. Because of this scattered system, students often miss deadlines or discover opportunities too late, even when they are interested and eligible to apply.

The Three Features at a Glance

Feature	Platform	Problem solved
1. Counsellor appointment booking	SETU	Email barrier to mental health access : 24–48 hr delay replaced with instant in-app booking
2. KGP doubt-solving platform	ApnaInsti	No searchable, course-indexed academic Q&A specific to IIT Kharagpur
3. Student resource & support hub	ApnaInsti	Scattered and unverified student resources; centralizes support and guidance through collaboration with students welfare group(SWG)

2.3 Counselor Appointment Booking on SETU

Problem

Booking a counselling session at SETU currently requires a student to compose and send a formal email, then wait up to 48 hours for confirmation. This two-step delay is both practically inconvenient and psychologically costly. People who want to attend a counselling session will just lose interest in getting help from someone in 24-48 hours of time.

Idea

A counsellor booking flow inside the SETU app's existing Wellness section. The student sees a list of available counselors with their focus areas and selects who that person wants to consult and that person picks a time slot, and receives instant confirmation. No email. No waiting. The booking is visible in their SETU app and can be rescheduled or cancelled directly, up to 6 hours before the session.

Key Features

- **Live availability calendar** showing open slots for the current and upcoming week
- **Instant booking confirmation** with session code; student identity hidden until session starts
- **Easy rescheduling/cancellation** with automated reminders (24 hrs & 1 hr prior)
- **Post-session follow-up prompt** to encourage continuity of care
- **Strict privacy**: SETU handles only scheduling; no clinical data stored

Implementation

Counsellor dashboard

A web-based panel for SETU staff. Each counsellor sets their weekly availability based on hours, session duration, blocked periods. This feeds directly into the student-facing booking view.

SETU app booking flow

A 'Book a Session' entry point added to the existing SETU app. Three-screen flow: browse counsellors, select a slot, confirm. The confirmation screen includes brief guidance on what to expect from a first session.

Notifications and privacy logic

Push notifications configured for confirmation, 24-hour reminder, 1-hour reminder, and post-session prompt. Student name hidden in counsellor view by default; revealed only when counsellor marks session active.

2.4 KGP Doubt Solving Platform on ApnaInsti

Problem

There is nowhere on the internet to find answers to academic doubts specific to IIT Kharagpur's courses, professors, and syllabi. WhatsApp groups are ephemeral. ERP has no Q&A. Generic platforms like Stack Exchange are not indexed by course codes or professor names. The result is that the same questions get asked every semester by every new batch and the answers, when they come, disappear again.

Idea

Introduce a doubt-solving platform within ApnaInsti, organized by course codes (e.g., MA11001, CS21003). Students can post questions, and peers, seniors, and TAs can respond. Answers are upvoted, and all content is permanently stored and searchable, creating a growing academic knowledge base for future batches.

Key Features

- **Course-based tagging** with full searchable archive of all Q&A
- **Community upvoting** to surface the most helpful answers
- **Verified TA responses** highlighted and prioritized
- **Semester tagging** for batch-relevant content
- **'Unanswered' filter** to direct attention to pending doubts
- **Secure login via KGP credentials** ensuring authenticity and zero spam

Implementation

Database and Search

Tables for questions, answers, votes, course tags, and user profiles. Full-text search index (PostgreSQL full-text or Meilisearch) configured across all content. Course code list populated from the KGP course catalogue.

Frontend on ApnaInsti

A 'Doubts' section added to the ApnaInsti navigation. Main view: filterable question feed by course code. Individual question pages show answers sorted by upvotes. Rich-text editor with LaTeX support for both questions and answers.

TA and Faculty Onboarding

TSG invites TAs from high-enrolment courses like Mathematics, Physics, core engineering to create verified accounts. Brief orientation on moderation and verification. Faculty participation optional but encouraged through department representatives.

Seed Content Launch

Senior students are invited to post five questions each from their most challenging courses, ensuring new users land on a platform that already has content. Announced via SETU notification and ApnaInsti homepage.

2.5 Student Resource & Support Hub on ApnaInsti

Problem

Student resources and guidance at IIT Kharagpur are currently scattered across multiple platforms such as WhatsApp groups, emails, and informal networks. Much of this information is unverified, inconsistent, or difficult to access when needed. As a result, students often struggle to find reliable guidance on academics, opportunities, and personal development, despite such resources existing within the campus ecosystem..

Idea

Create a centralized Resource & Support Hub on ApnaInsti in collaboration with the Students' Welfare Group (SWG), where students can easily find verified information on academics, opportunities, and general support. This brings all useful resources into one place, making them easy to access and reliable.

Key Features

- Curated and verified resources across academics, career opportunities, and well-being
- Dedicated sections for internships, competitions, research, and skill-building
- Guidance articles and FAQs contributed by SWG and experienced students
- Categorized and searchable content for easy navigation
- Access restricted to KGP students via existing login system

Implementation

Content and Collaboration

Tables for questions, answers, votes, course tags, and user profiles. Full-text search index configured across all content. Course code list populated from the KGP course catalogue.

Platform Integration

A “Resources” section is added to ApnaInsti with a simple, organized interface. Content is displayed in categorized formats with search and filter options.

Content Management System

A lightweight backend system allows SWG members to add, update, and review resources regularly, ensuring information remains accurate and useful.

Launch Strategy

Initial content is seeded with commonly required resources and guides. The feature is introduced through ApnaInsti announcements and campus-wide communication to ensure awareness and adoption.

2.6 Expected Outcomes

For Students — Immediate

- Access restricted to KGP students via existing login system Counselling sessions can be booked in under two minutes, anytime, without email delays
- Academic doubts are permanently answered and searchable, avoiding repeated questions
- Easy access to verified resources and support through a centralized hub in collaboration with SWG

For the Institution

- Increased counselling uptake due to reduced process friction
- A continuously growing academic knowledge base specific to IIT Kharagpur
- Structured and reliable dissemination of student resources through SWG collaboration

2.7 Conclusion

These features remove key barriers between students and support, answers, and opportunities. They simplify access without changing student intent. No new platforms are required, only enhancements to SETU and ApnaInsti.

Proposal 3

Establishing a Unified Digital Ecosystem for Research and Project Collaboration

3.1 Goals

To bridge the communication gap between faculty and students by centralizing all campus research opportunities, term projects, and lab openings into a single, verified digital platform. The program aims at a data-driven recruitment system that promotes meritocracy, transparency, and interdisciplinary collaboration. By providing a standardized application process, the initiative seeks to ensure every student has equitable access to high-impact research, regardless of their personal networks.

3.2 Overview and Current Scenario

Currently, the dissemination of research opportunities and project openings at the institute is highly decentralized, leading to several systemic inefficiencies:

- **Fragmented Communication:** Information is scattered across departmental emails, physical notice boards, and informal social media groups, causing many students to miss critical deadlines.
- **Manual Screening Burdens:** Professors often receive a high volume of unorganized emails and CVs, making it difficult to filter candidates based on specific technical skills or academic standing.
- **Lack of Tracking:** There is no institutional record of student participation in non-credited projects, making it difficult for students to validate their experience during CDC internship procedures.

3.3 Proposed Solutions and Their Impact

a. Faculty-Centric Management Dashboard

To streamline the recruitment of student researchers, the portal will provide professors with a dedicated management suite.

- **Easy Posting:** A simplified interface for posting project requirements, required tech stacks, and duration.
- **Automated Filtering:** Tools to sort applicants by GPA, specific course completions (e.g., Quantum Mechanics), or validated skills.
- **Team Management:** A centralized dashboard to track active student members, project milestones, and past contributors.

b. Student Opportunity Hub

A comprehensive interface designed to empower students in their academic and professional pursuits.

- **Searchable Database:** A real-time repository of open roles across all departments, filterable by interest areas like AI, Core Engineering, or Social Sciences.
- **Follow Notifications:** A "Follow Lab" feature that sends instant alerts to students when their preferred research groups post new openings.
- **Digital CV Profile:** A standardized digital profile that pulls verified academic data directly from the ERP, ensuring credibility and reducing the 15-20 minute manual paperwork typically required for applications.

c. Secure Verification and Integration

To maintain the integrity of the platform, the system will utilize the institute's existing digital infrastructure.

- **SSO Authentication:** Integration with the Institute ERP/G-Suite to ensure only currently enrolled students and active faculty can access the portal.
- **Role-Based Access:** Automatic verification of departmental affiliations to suggest relevant local opportunities while allowing for inter-departmental exploration.

d. Streamlining Project Allocation for UG and PG Students

- **Centralized BTP/MTP listings** to eliminate scattered and unfiltered information
- **Clear project descriptions** including prerequisites, domains, and expectations
- **Standardized application process** for BTP and MTP selections
- **Better student-project alignment** based on interests and verified profiles
- **Efficient faculty selection** through structured and comparable applications

3.4 Benefits to the Institution

- **Enhanced Research Output:** By matching the right talent to the right projects faster, the institute can accelerate the pace of publications and patent filings.
- **Improved Placement Rates:** Standardized project experience and verified digital CVs will better prepare students for competitive CDC internship and placement cycles.
- **Data-Driven Insights:** The administration will gain valuable metrics on campus engagement, popular research trends, and departmental resource needs.